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MCY 612 Cloud Security Lab#6: AWS GuardDuty

In this lab, you will study the Amazon GuardDuty. You will learn how to use the AWS CLI to protect your AWS account with intelligent threat detection.

Tools and Materials

AWS credentials
AWS CloudShell CLI

Note: You need to replace words in *italics* with your information.

Start your AWS CloudShell terminal. Create an Amazon GuardDuty detector with the following command. A detector is a resource that represents the GuardDuty service. To start using GuardDuty, you must create a detector where you enable the service.

```
aws guardduty create-detector --enable
```

What is the complete output?

```
~ $ aws guardduty create-detector --enable
{
  "DetectorId": "dccb3cf6e86907aeccf259e9c0087e24"
}
```

List the detector in your AWS account with the following command:

```
aws guardduty list-detectors
```

What is the complete output? What is the DetectorId?

```
~ $ aws guardduty list-detectors
{
  "DetectorIds": [
    "dccb3cf6e86907aeccf259e9c0087e24"
  ]
}
```

Retrieve the Amazon GuardDuty detector with the following command:

```
aws guardduty get-detector --detector-id the_DetectorId
```

What is your command and its complete output?

```
~ $ aws guardduty get-detector --detector-id dccb3cf6e86907aeccf259e9c0087e24
```

```
{
  "CreatedAt": "2025-04-28T01:05:51.100Z",
  "FindingPublishingFrequency": "SIX_HOURS",
  "ServiceRole": "arn:aws:iam::881490118823:role/aws-service-role/guardduty.amazonaws.com/AWSServiceRoleForAmazonGuardDuty",
  "Status": "ENABLED",
  "UpdatedAt": "2025-04-28T01:05:51.100Z",
  "DataSources": {
    "CloudTrail": {
      "Status": "ENABLED"
    },
    "DNSLogs": {
      "Status": "ENABLED"
    },
    "FlowLogs": {
      "Status": "ENABLED"
    },
    "S3Logs": {
      "Status": "ENABLED"
    },
    "Kubernetes": {
      "AuditLogs": {
        "Status": "ENABLED"
      }
    },
    "MalwareProtection": {
      "ScanEc2InstanceWithFindings": {
        "EbsVolumes": {
          "Status": "ENABLED"
        }
      }
    }
  }
}
```

```
    },
    "ServiceRole": "arn:aws:iam::881490118823:role/aws-service-role/malware-protection.guardduty.amazonaws.com/AWSServiceRoleForAmazonGuardDutyMalwareProtection"
  },
  "Tags": {},
  "Features": [
    {
      "Status": "ENABLED",
      "UpdatedAt": "2025-04-28T01:06:57+00:00"
    },
    {
      "Name": "FLOW_LOGS",
      "Status": "ENABLED",
      "UpdatedAt": "2025-04-28T01:06:57+00:00"
    },
    {
      "Name": "S3_DATA_EVENTS",
      "Status": "ENABLED",
      "UpdatedAt": "2025-04-28T01:05:51+00:00"
    },
    {
      "Name": "EKS_AUDIT_LOGS",
      "Status": "ENABLED",
      "UpdatedAt": "2025-04-28T01:05:51+00:00"
    },
    {
      "Name": "EBS_MALWARE_PROTECTION",
      "Status": "ENABLED",

```

```
    "UpdatedAt": "2025-04-28T01:05:51+00:00"
  },
  {
    "Name": "RDS_LOGIN_EVENTS",
    "Status": "ENABLED",
    "UpdatedAt": "2025-04-28T01:05:51+00:00"
  },
  {
    "Name": "EKS_RUNTIME_MONITORING",
    "Status": "DISABLED",
    "UpdatedAt": "2025-04-28T01:05:51+00:00",
    "AdditionalConfiguration": [
      {
        "Name": "EKS_ADDON_MANAGEMENT",
        "Status": "DISABLED",
        "UpdatedAt": "2025-04-28T01:05:51+00:00"
      }
    ]
  },
  {
    "Name": "LAMBDA_NETWORK_LOGS",
    "Status": "ENABLED",
    "UpdatedAt": "2025-04-28T01:05:51+00:00"
  },
  {
    "Name": "RUNTIME_MONITORING",
    "Status": "DISABLED",
    "UpdatedAt": "2025-04-28T01:05:51+00:00",
    "AdditionalConfiguration": [
```

```
{
  "Name": "EKS_ADDON_MANAGEMENT",
  "Status": "DISABLED",
  "UpdatedAt": "2025-04-28T01:05:51+00:00"
},
{
  "Name": "ECS_FARGATE_AGENT_MANAGEMENT",
  "Status": "DISABLED",
  "UpdatedAt": "2025-04-28T01:05:51+00:00"
},
{
  "Name": "EC2_AGENT_MANAGEMENT",
  "Status": "DISABLED",
  "UpdatedAt": "2025-04-28T01:05:51+00:00"
}
]
}
```

Create an S3 bucket to host a test file.

`aws s3api create-bucket --bucket your_username-guardduty-test`

What is your command and its complete output?

```
~ $ aws s3api create-bucket --bucket carrd12-guardduty-test
{
  "Location": "/carrd12-guardduty-test"
}
~ $ █
```

Create a file called test.html with the following content.

```
<html><body><h1>Amazon S3</h1>
Hello World!
</body></html>
```

Upload the file to the bucket via the following command

`aws s3 cp test.html s3://your_username-guardduty-test`

What is your command and its complete output?

```
~ $ aws s3 cp test.html s3://carrd12-guardduty-test
upload: ./test.html to s3://carrd12-guardduty-test/test.html
~ $ █
```

Figure out the public IP address of your CloudShell with the following command:

```
curl ipinfo.io/ip
```

curl is a command line tool for getting or sending data including files using URL syntax and HTTP/HTTPS.
ipinfo.io is a tool to show your IP Address.

What is your command and its complete output? What are the first three octets of your IP address? For example, the IP address that I got was 35.173.243.183. The first three octets of my IP address is 35.173.243

```
~ $ curl ipinfo.io/ip
3.82.96.235~ $
```

To simulate a threat, use the CIDR (Classless Inter-Domain Routing) subnet to represent known malicious IP addresses. To do this, create a text file named threatlist.txt with the following content:

`the_first_three_octets_of_your_IP_address.0/24`

Create an S3 bucket to store the threat list for GuardDuty.

```
aws s3api create-bucket --bucket your_username-guardduty
```

What is your command and its complete output?

```
~ $ aws s3api create-bucket --bucket carrd12-guardduty
{
  "Location": "/carrd12-guardduty"
}
~ $
```

Put the threat list into the bucket with the following command.

```
aws s3api put-object --bucket your_username-guardduty --key threatlist.txt --body threatlist.txt
```

What is your command and its complete output?

```
~ $ aws s3api put-object --bucket carrd12-guardduty --key threatlist.txt --body threatlist.txt
{
  "ETag": "\"0c526e62298c3ed415785d279857f1ab\"",
  "ChecksumCRC64NVME": "00Pqvg64/wM=",
  "ChecksumType": "FULL_OBJECT",
  "ServerSideEncryption": "AES256"
}
~ $
```

Create a new ThreatIntelSet with the following command. A ThreatIntelSet consists of known malicious IP addresses.

```
aws guardduty create-threat-intel-set --detector-id your_DetectorId --name customThreatSet1 --format TXT
--location s3://your_username-guardduty/threatlist.txt --activate
```

What is your command and its complete output?

```
~ $ aws guardduty create-threat-intel-set --detector-id dccb3cf6e86907aecf259e9c0087e24 --name customThreatSet1 --format TXT --location s3://carrd12-guardduty/threatlist.txt --activate
{
  "ThreatIntelSetId": "8ccb3cfca93d7cc70e78403c07d5ff7e"
}
```

List the ThreatIntelSets of the GuardDuty service with the following command:

aws guardduty list-threat-intel-sets --detector-id **your_DetectorId**

What is your command and its complete output? What is the ThreatIntelSetId of customThreatSet1?

"ThreatIntelSetIds": "8ccb3cfca93d7cc70e78403c07d5ff7e"

```
~ $ aws guardduty list-threat-intel-sets --detector-id dccb3cf6e86907aecf259e9c0087e24
{
  "ThreatIntelSetIds": [
    "8ccb3cfca93d7cc70e78403c07d5ff7e"
  ]
}
```

Run the following command to retrieve the ThreatIntelSet that is specified by the ThreatIntelSet ID:

aws guardduty get-threat-intel-set --detector-id **the_DetectorId** --threat-intel-set-id **the_ThreatIntelSetId**

What is your command and its complete output?

```
~ $ aws guardduty get-threat-intel-set --detector-id dccb3cf6e86907aecf259e9c0087e24 --threat-intel-set-id 8ccb3cfca93d7cc70e78403c07d5ff7e
{
  "Name": "customThreatSet1",
  "Format": "TXT",
  "Location": "s3://carrd12-guardduty/threatlist.txt",
  "Status": "ACTIVE",
  "Tags": {}
}
```

Use your CloudShell terminal, which is now a blacklisted terminal, to access an S3 bucket. Run the following command to download a webpage:

aws s3 cp s3://**your_username**-guardduty-test/test.html .

This command is invoked from an IP address that is included on a threat list. GuardDuty will detect this potentially malicious activity and generate a finding with the UnauthorizedAccess:S3/MaliciousIPCaller.Custom type.

Wait for 5 – 15 minutes and then display a list of all findingIds that match the UnauthorizedAccess:S3/MaliciousIPCaller.Custom finding type with the following command:

aws guardduty list-findings --detector-id **your_DetectorId** --finding-criteria '{"Criterion":{"type":{"Eq":["UnauthorizedAccess:S3/MaliciousIPCaller.Custom"]}}}'

What is your command and its complete output? What is the first FindingId?

"FindingIds": ["4ecb3cfe69395403c99cd718b7582161"]

```
~ $ aws guardduty list-findings --detector-id dccb3cf6e86907aecf259e9c0087e24 --finding-criteria '{"Criterion":{"type":{"Eq":["UnauthorizedAccess:S3/MaliciousIPCaller.Custom"]}}}'
{
  "FindingIds": [
    "4ecb3cfe69395403c99cd718b7582161"
  ]
}
```

Describe one GuardDuty finding with the following command:

aws guardduty get-findings --detector-id **your_DetectorId** --finding-id **the_first_FindingId**

What is your command and its complete output? What is the Description of the finding? What is the RemoteIpDetails? What is Severity? What is the Title?

"Description": "An API was used to access a bucket from an IP address on a custom threat list.",
 "Severity": 8.0,

"Title": "The API GetObject was invoked from an IP address on a custom threat list.",

```
~ $ aws guardduty get-findings --detector-id dccb3cf6e86907aeccf259e9c0087e24 --finding-id 4ecb3cfe69395403c99cd718b7582161
{
  "Findings": [
    {
      "AccountId": "881490118823",
      "Arn": "arn:aws:guardduty:us-east-1:881490118823:detector/dccb3cf6e86907aeccf259e9c0087e24/finding/4ecb3cfe69395403c99cd718b7582161",
      "CreatedAt": "2025-04-28T01:22:14.258Z",
      "Description": "An API was used to access a bucket from an IP address on a custom threat list.",
      "Id": "4ecb3cfe69395403c99cd718b7582161",
      "Partition": "aws",
      "Region": "us-east-1",
      "Resource": {
        "AccessKeyDetails": {
          "AccessKeyId": "AKIA42PHHZCTQDSFLZ02",
          "PrincipalId": "AIDA42PHHZCTSJOBGHHJW",
          "UserName": "carrd12-admin-user",
          "UserType": "IAMUser"
        },
        "S3BucketDetails": [
          {
            "Arn": "arn:aws:s3:::carrd12-guardduty-test",
            "Name": "carrd12-guardduty-test",
            "Type": "Destination",
            "CreatedAt": "2025-04-28T01:10:07+00:00",
            "Owner": {
              "Id": "1ee16e5a4c54d5e3c3dc09019713274b27e3668a7c732ea1337808cd57ce0d36"
            },
            "DefaultServerSideEncryption": {
              "EncryptionType": "AES256"
            }
          }
        ]
      }
    }
  ]
}
```

AWS CloudShell

```
},
"PublicAccess": {
  "PermissionConfiguration": {
    "BucketLevelPermissions": {
      "AccessControllist": {
        "AllowsPublicReadAccess": false,
        "AllowsPublicWriteAccess": false
      },
      "BucketPolicy": {
        "AllowsPublicReadAccess": false,
        "AllowsPublicWriteAccess": false
      },
      "BlockPublicAccess": {
        "IgnorePublicAcls": true,
        "RestrictPublicBuckets": true,
        "BlockPublicAcls": true,
        "BlockPublicPolicy": true
      }
    },
    "AccountLevelPermissions": {
      "BlockPublicAccess": {
        "IgnorePublicAcls": false,
        "RestrictPublicBuckets": false,
        "BlockPublicAcls": false,
        "BlockPublicPolicy": false
      }
    }
  },
  "EffectivePermission": "NOT_PUBLIC"
}
```

```
        "EffectivePermission": "NOT_PUBLIC"
      }
    ],
    "ResourceType": "S3Bucket"
  },
  "SchemaVersion": "2.0",
  "Service": {
    "Action": {
      "ActionType": "AWS_API_CALL",
      "AwsApiCallAction": {
        "Api": "GetObject",
        "CallerType": "Remote IP",
        "RemoteIpDetails": {
          "City": {
            "CityName": "Ashburn"
          },
          "Country": {
            "CountryName": "United States"
          },
          "GeoLocation": {
            "Lat": 39.0469,
            "Lon": -77.4903
          },
          "IpAddressV4": "3.82.96.235",
          "Organization": {
            "Asn": "14618",
            "AsnOrg": "AMAZON-AES",
            "Isp": "Amazon.com",
```



```
        "Org": "Amazon.com"
      }
    },
    "ServiceName": "s3.amazonaws.com",
    "AffectedResources": {}
  }
},
"Evidence": {
  "ThreatIntelligenceDetails": [
    {
      "ThreatListName": "customThreatSet1",
      "ThreatNames": [
        "Customer Threat Intel"
      ]
    }
  ]
},
"Archived": false,
"Count": 2,
"DetectorId": "dccb3cf6e86907aeccf259e9c0087e24",
"EventFirstSeen": "2025-04-28T01:17:05.000Z",
"EventLastSeen": "2025-04-28T01:23:09.000Z",
"ResourceRole": "TARGET",
"ServiceName": "guardduty",
"AdditionalInfo": {
  "Value": "{\ authenticationMethod\": \"AuthHeader\", \"threatListName\": \"customThreatSet1\", \"threatName\": \"Customer Threat Intel\" }",
  "Type": "default"
}
},
}
~ $ █
```

```
    "Severity": 8.0,
    "Title": "The API GetObject was invoked from an IP address on a custom threat list.",
    "Type": "UnauthorizedAccess:S3/MaliciousIPCaller.Custom",
    "UpdatedAt": "2025-04-28T01:28:16.216Z"
  }
]
}
~ $ █
```

Cleanup

Delete the GuardDuty detector:

aws guardduty delete-detector --detector-id **your_DetectorId**

What is your command?

```
~ $ aws guardduty delete-detector --detector-id dccb3cf6e86907aeccf259e9c0087e24
```

Delete both S3 Buckets.

What are your commands and their complete outputs?

```
~ $ aws s3 rm s3://carrd12/test.html
delete: s3://carrd12/test.html
```

```
~ $ aws s3 rb s3://carrd12-guardduty --force
delete: s3://carrd12-guardduty/threatlist.txt
remove_bucket: carrd12-guardduty
```

Submission instructions

After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

The purpose of this lab was to investigate and put into practice Amazon GuardDuty, which uses intelligent threat detection to improve the security of an AWS account. The goal was to comprehend how GuardDuty detects possible security threats using threat information, anomaly detection, and machine learning. I used the AWS CLI to enable GuardDuty, set it up to monitor my AWS environment, and examined the results it produced in order to finish the lab. I learned how to set up automated security monitoring and properly handle suspicious activity thanks to this practical experiment.